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Interactions of deprotonated phenylalanine with gold Clusters: Theoretical study with prospects for amino acid detection

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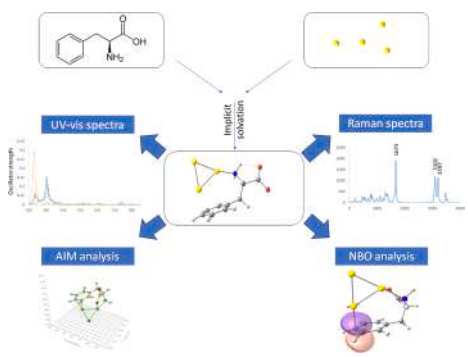
HIGHLIGHTS

- Interactions of phenylalanine (Phe) and Au clusters (NCs) have been studied using theory for the first time.
- The NCs with magic number of electrons were the most favorable ones.
- Metal-ligand intracomplex interactions were partially covalent and partially electrostatic.
- The presence of gold significantly enhanced the Raman spectra signal of Phe.

GRAPHICAL ABSTRACT



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ABSTRACT

Nanomaterials are widely used nowadays in industry and medicine. The specific properties of gold nanoclusters (Au NCs) are chemical stability, low cytotoxicity, low photobleaching, high sensitivity to the molecular environment. This set of properties allows to use Au NCs as nanosensors in bioimaging and diagnostics. We have investigated gold cluster complexes with proteinogenic amino acid phenylalanine (Phe). Detection of phenylalanine is essential for diagnostics of phenylketonuria, vitiligo, sclerosis, cancer, tuberculosis, etc. We have studied the complexes of Phe with Au_n^q clusters with atomic number equal 1–6, 8, 20 and a charge equal 0–2. We have established that the clusters Au_4^0, Au_2^{1+} and Au_3^{2+} form the most stable complexes with Phe among NCs with charge 0, +1 and + 2, respectively. Intracomplex interactions have been studied using Atoms-In-Molecules (AIM) theory and Natural Bond Orbital (NBO) analysis. It has been shown that metal–ligand intracomplex interactions are partially covalent and partially electrostatic. Also, we have simulated the UV–vis absorption and Raman spectra of the Phe-Au NCs. We have established that the clusters possess prospective features if being used for colorimetric and Raman detection of Phe. Au_{20} cluster is remarkable for its six-times enhancement of the Raman signal. Moreover, our study provides insights into metal–ligand interactions for clusters synthesized inside a polypeptide globula. Hence, to the best of our knowledge this is a first attempt to perform a detailed analysis of Phe interactions with gold using quantum chemical calculations.

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1. Introduction

Nanomaterials are widely used nowadays in many areas of industry, medicine, food production, textiles, agriculture [1], etc. Coinage metal nanoclusters (NCs) are sub-nanometer size objects (<2 nm) with excellent physicochemical characteristics and tunable optical properties. The specific properties of gold nanoclusters are low cytotoxicity, high water solubility, photostability, size-tunable emission maxima, large Stokes shift and biocompatibility [2]. Among the gold clusters those that follow magic numbers (2, 10, 18, 36, 54, and 86) possess the most attractive properties. This «superatom» concept has been developed by Hannu Hakkinen and collaborators in 2008 [3]. According to this concept, the gold core is a sort of giant atom which is stabilized when its number of electrons and electron-accepting groups follows certain magic numbers. In particular, the original formula for a number n* is following:

$$n^* = N_{Au} - W - q \quad (1)$$

where N_{Au} is the amount of Au atoms, or the number of unpaired electrons, W is the number of electron-accepting groups, whereas q is the charge of the nanocluster [4].

Gold NCs are used to be synthesized both in vacuum and in aqueous solutions. In the vacuum, the clusters are synthesized in the isolated state, whereas in solution biopolymers and small ligands allow to prevent self-aggregation of the clusters. The usage of ligands during the development and synthesis of Au NCs allows high biocompatibility which is significant for biosensing, bioimaging and other biomedical applications. Gold clusters are often stabilized by aromatic amino acid ligands [5–7]. In particular, phenylalanine (Phe) is used for the gold cluster synthesis as a stabilizer, a reducing and a capping agent [8]. Moreover, gold nanoparticles can be used for colorimetric detection of Phe [9,10]. However, to the best of our knowledge our study is a first attempt to perform theoretical calculations of the Phe-Au interactions.

Phenylalanine is a biomarker of phenylketonuria [11], breast cancer [12], lung cancer and tuberculosis [13], etc. Moreover, Phe is known to be a biomarker of whole body proteolysis; the chronically elevated level of serum Phe is cardiotoxic and leads to the development of coronary artery disease, heart failure, and cardiac fibrosis [14]. For this reason, the development of novel precise methods of Phe detection are in great demand.

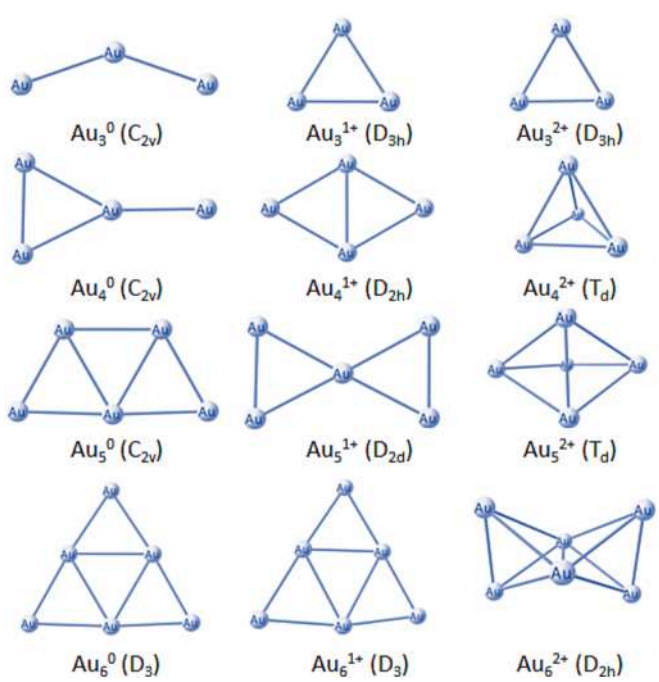

Fig. 1. The most stable isomers for isolated Au_n^q (n = 3–6, q = 0–2) nanoclusters.

Application of metal nanoclusters and nanoparticles is in line with the most recent trends in bioanalytical chemistry for precise determination of amino acids in biological fluids [15,16]. In this case, amino acid detection can be done via colorimetric analysis, fluorescence spectroscopy, infrared and Raman spectroscopy [17]. The deprotonated form of Phe is of particular interest because (1) deprotonated amino acids possess higher Gibbs binding energy with gold clusters [18]; (2) some experiments show that alkaline pH is favorable for gold nanoparticle synthesis [19]. In this paper, we have regarded Phe at a fully deprotonated form which corresponds to pH > 9.1.

Recently, there has been a growing attention on studying the nature of interactions between gold clusters and various proteinogenic amino acids [18,20,21]. In a recent paper by Peng and co-authors [22] binding of 20 proteinogenic amino acids and ultrasmall Au_3 cluster was studied using machine learning; in particular, they showed that the main binding site of Phe is the nitrogen atom. The research on the most favorable amino acid residues and their binding to gold is necessary since it is significant for the development of a theoretical viewpoint on how the amino acid-Au binding affects optical properties and Raman spectra of gold nanoclusters. To understand the interplay of gold clusters and amino acids is crucial for the development and design of tiny biosensors [23]. Next to cysteine and thiols, aromatic amino acids are the most frequently utilized capping agent for the experimental development of gold nanoclusters [6,7,24]. The pioneering works on Ag/Au cluster-aromatic amino acid hybrids were performed by Bonacic-Koutecky's group [25,26]. In particular, they found that the enhancement of absorption of amino acids/peptides can be achieved by ultra-small silver clusters due to the interaction of intense intracluster excitations with the π- π* excitations of chromophoric amino acids [27].

The Raman and surface-enhanced Raman spectra (SERS) of phenylalanine have been published multiple times previously [28,29]. However, despite of several studies on gold nanoparticles interactions with phenylalanine [30,31], a detailed depiction of Phe-Au NC Raman spectra is missing. In this paper we had an aim to find the most stable complexes of Phe^{-1} with gold NCs. The adsorption/desorption of benzol moiety of the main aromatic amino acid Phe (precursor for the biosynthesis of all aromatic amino acids) and gold using nanoclusters containing up to six atoms have been used as a model system. To do so, we systematically evaluated the Gibbs energy of complexation between Phe and gold NCs. Next, we have studied intracomplex interactions using topological wavefunction analysis. Also, we have simulated the UV–vis absorption and Raman spectra of the Phe-Au complexes to evaluate the potential of these methods for Phe detection in the aqueous solutions *in vitro*.

2. Computational details

The Phe-Au complexes have been studied using density functional theory (DFT), in particular PBE functional which is known to perform precise calculations for organometallic complexes of gold [32]. Since DFT functionals do not take electron dispersion into account we also used Grimme dispersion correction D3(BJ) [33,34]. Def2-TZVP basis set [35] was used along with effective core potential (ECP) def2-ECP for Au atoms [36]. Thus, PBE-D3(BJ)/def2-TZVP method was used for geometry optimization and hessian calculation. This method was applied to establish Gibbs binding energy for the complexes according to complexation equation:

$$\mathrm{Phe} + \mathrm{Au_n}^q \rightarrow \mathrm{Phe} - \mathrm{Au_n}^q$$

Then, the Gibbs energy of complexation (ΔG°) formula has been used as following:

$$\Delta G^\circ = G^\circ_{Au} + G^\circ_{Phe} - G^\circ_{complex}$$

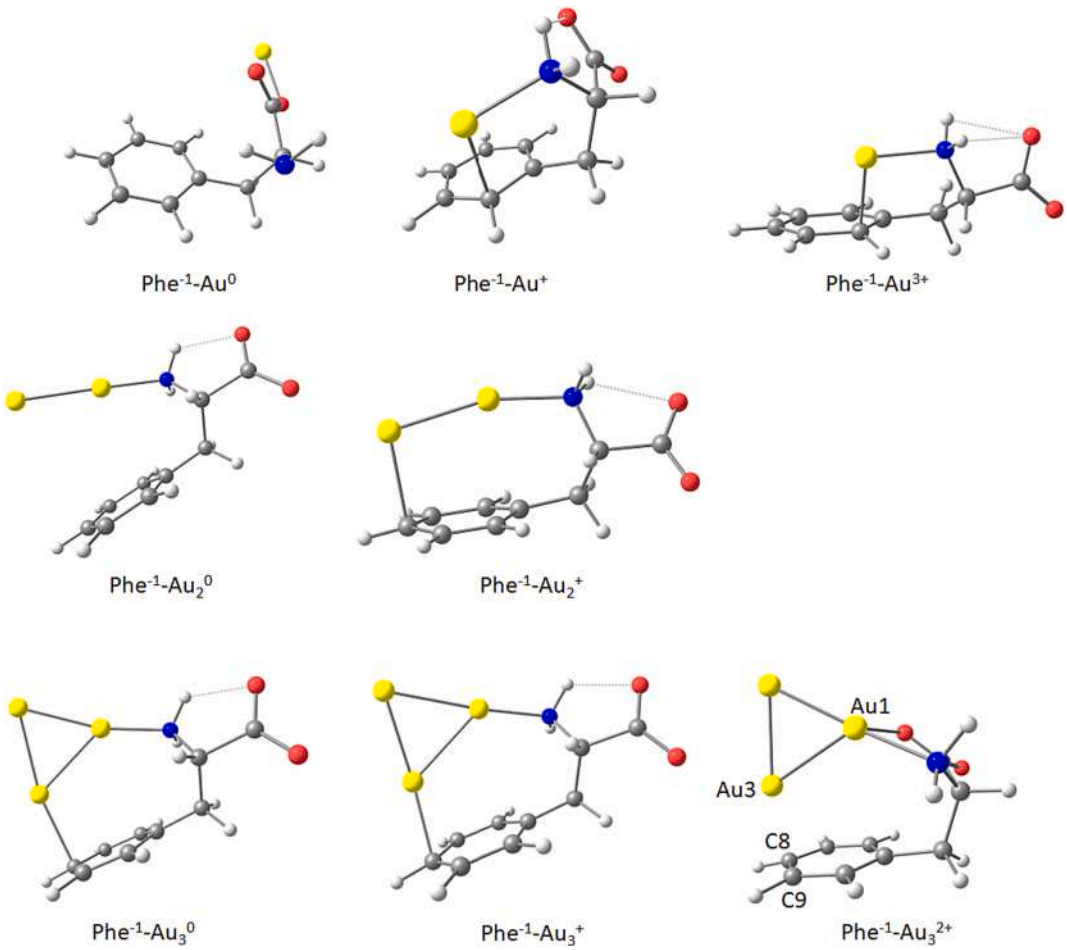

Fig. 2. The most stable complexes of gold nanoclusters with Phe^{-1} (PBE-D3 method).

where ΔG°_{Au} is a Gibbs energy of a gold cluster, G°_{Phe} is a Gibbs energy of Phe, $G^{\circ}_{complex}$ is a Gibbs energy of the Phe-Au^{q}_{n} complex. Water was taken into account using the CPCM model implemented in Orca 5.0.2 [37].

We have examined intracomplex organometallic interactions using Bader's quantum theory «Atoms-In-Molecules» (QTAIM) [38] and PBE-D3/def2-TZVP method. The QTAIM analysis has been done using Multiwfn 3.6 programme [39].

UV–vis absorption spectra have been simulated using M062X/def2-TZVP and CAM-B3LYP methods. M062X approved its efficacy on absorption spectra of transition metals [40], organics [41], and organometallic complexes [42,43]. CAM-B3LYP is known to perform correct TDDFT calculation for gold complexes with organic ligands [44].

Also, we have simulated Raman spectra of Phe complexes with gold using B3LYP-D3(BJ). B3LYP has demonstrated precise calculations on similar systems [45,46]. Natural bond orbital (NBO) analysis was performed to establish the nature of the major transitions in the Raman spectra.

3. Results and discussion

3.1. Isolated gold clusters

First, we have established the most stable isomers of the isolated clusters up to six atoms in size.

The Au^{0}_{3} cluster has a triangle shape (Fig. 1) with an obtuse valence angle equal to 141.3° and Au-Au bond length equal to 2.565 Å according to PBE-D3. This is in agreement with previous studies [47]. Triatomic clusters with charge + 1 and + 2 are equilateral triangles with a side equal to 2.633 Å and 2.569 Å for Au^{1+}_{3} and Au^{2+}_{3} clusters, respectively.

The shape of tetramer clusters largely depends on charge. For example, a T-shaped Au^{0}_{4} cluster with a C_{2v} symmetry has the lowest absolute energy as compared to other isomers. This is not supported by the earlier studies in the gas phase [47,48], the reason is probably CPCM water which was used in our case. Indeed, the potential energy difference between the rhombus and T-shaped isomer of the Au^{0}_{4} is equal zero in the gas phase experiment [49]. In our case in presence of PCM water, the T-shape isomer is 4.4 kcal mol^{-1} more stable that the rhombus. However, Au^{1+}_{4} is a rhombus (D_{2h}) with a side equal to 2.657 Å, whereas Au^{2+}_{4} is a tetraeder (T_d) with a side equal to 2.714 Å.

The most stable isomer of Au^{0}_{5} is a trapezoid with C_{2v} symmetry which is in agreement with previous theoretical [47] and experimental studies [49]. The D_{2d} pentamer is the most stable form of Au^{1+}_{5}, whereas Au^{2+}_{5} cluster possesses T_d symmetry. Au^{0}_{6} and Au^{1+}_{6} hexamers both have a D_3 symmetry, whereas Au^{2+}_{6} is a D_{2h}. The data for the neutral and cationic clusters are in line with previous studies [47,50].

Therefore, clusters with charge 0 and + 1 remain planar up to 6 atoms size, whereas Au^{2+}_{n} clusters transform from 2D to a 3D structure at n = 4. All geometries of the depicted clusters are in line with previous studies. Au_8 cluster possesses a planar D_{4h} symmetry, whereas Au_{20} is a T_d tetrahedral structure.

3.2. Gold clusters in complex with deprotonated phenylalanine

We have analyzed the geometries of the clusters optimized in complex with Phe^{-1}. For example, neutral gold atom attaches solely to the

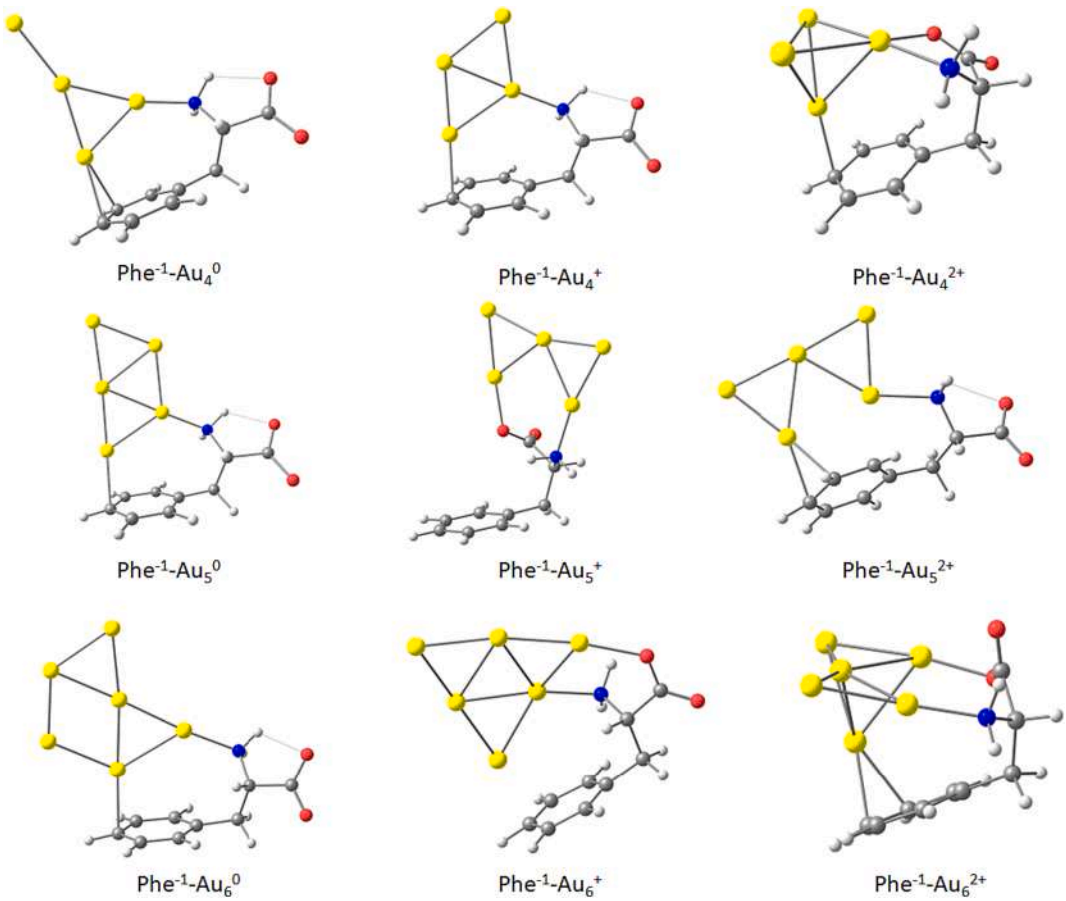

Fig. 2. (*continued*).

carboxylate forming a monodentate complex (Fig. 2). Au^{+} ion forms bonds with both phenyl and amino group of Phe^{-1} and forms a bidentate complex. Au^{3+} ion also binds to the amino group and phenyl ring simultaneously.

Au_2^0 forms bonds with the amino group, whereas Au_2^{+} forms bonds with both amino group and phenyl. Among three-atomic clusters, only Au_3^{2+} forms bonds with all three functional groups of Phe^{-1}: amino group, carboxylate and phenyl. Au_3^{1+} and Au_3^0 form bonds with phenyl and amino group ignoring the carboxylate. The Au_3^0 cluster changes its symmetry from C_{2v} to D_{3h} upon the addition of Phe^{-1}.

Four-atomic clusters do not change their symmetry upon the addition of phenylalanine which results in production of C_{2v}, D_{2h} and T_d symmetry clusters of Phe^{-1}-Au_4^0, Phe^{-1}-Au_4^{1+} and $Phe^{-1}Au_4^{2+}$ systems, respectively. Au_4^0 and Au_4^{1+} clusters form bidentate complexes, whereas the Au_4^{2+} cluster forms a tridentate system with Phe^{-1}.

All five-atomic clusters possess C_{2v} symmetry in complex with Phe. Thus, Au_5^{1+} transforms from D_{2d} to C_{2v}, whereas Au_5^{2+} changes its symmetry from T_d to C_{2v} upon the addition of Phe. Au_5^0 attaches to phenyl

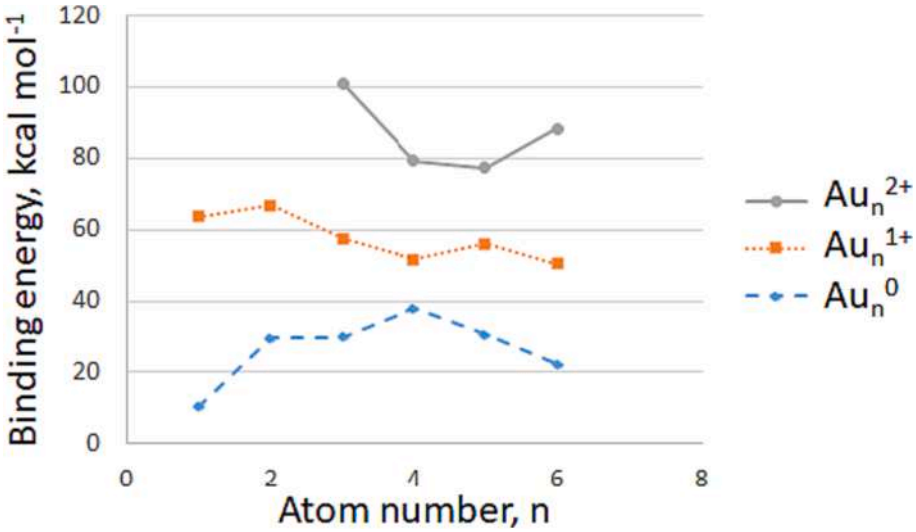

Fig. 3. Intracomplex binding energies according to the PBE-D3/def2-TZVP method.

Table 1
Topology analysis data (in hartree) according to Bader's QTAIM and PBE-D3 method.

Complex	BCP	ρ(r)	$\nabla^2\rho(r)$	G(r)	V(r)	E(r)	E_{bond}, kcal/mol
Phe^{-1}- Au_3^{2+}	Au-N	0.1243	0.4155	0.1522	−0.2004	−0.0483	62.9
Phe^{-1}- Au_3^{2+}	Au-O	0.1091	0.4245	0.1411	−0.1760	−0.349	55.2
Phe^{-1}- Au_3^{2+}	Au-C	0.0943	0.2255	0.0877	−0.1191	−0.0314	37.4
Phe^{-1}- Au_2^{1+}	Au-N	0.1339	0.4719	0.1716	−0.2252	−0.0536	70.7
Phe^{-1}- Au_2^{1+}	Au-C	0.0809	0.1940	0.0715	−0.0945	−0.0230	29.6
Phe^{-1}- Au_4^{0}	Au-N	0.1241	0.4522	0.1592	−0.2053	−0.0461	64.4
Phe^{-1}- Au_4^{0}	Au-C	0.0847	0.2307	0.0812	−0.1047	−0.0235	32.9

and amino group ignoring the carboxylate. Au_5^{1+} attaches to carboxylate and amino group ignoring the aromatic ring. The Au_5^{2+} cluster ignores the carboxylate forming bonds with phenyl and the amino group.

The Au_6^{2+} cluster changes its symmetry to C_s upon the addition of Phe^{-1}. This structure forms bonds with all the functional groups of the amino acid. Au_6^{1+} is also tridentate, whereas Au_6^{0} is bidentate and ignores

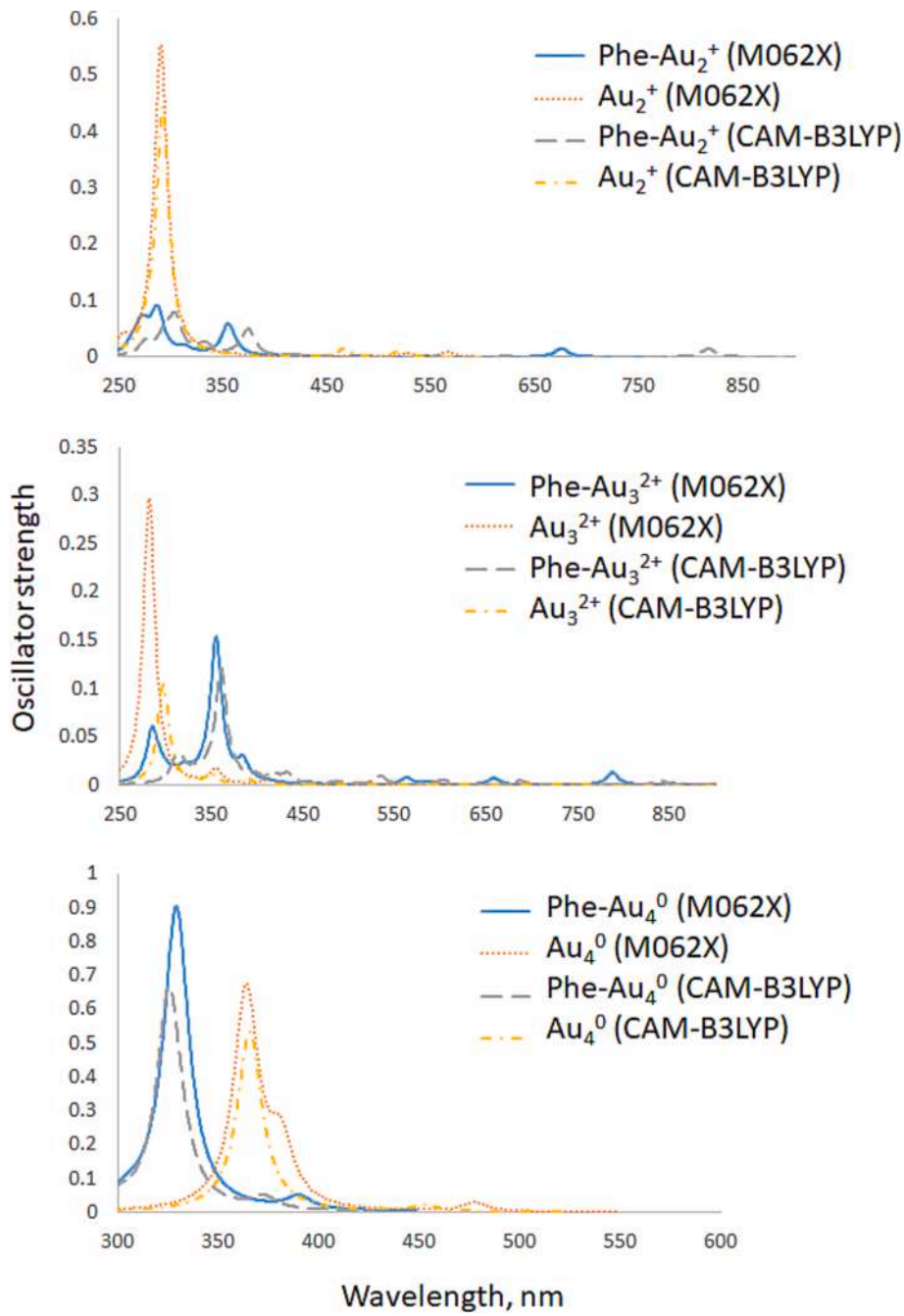

Fig. 4. Simulated absorption spectra of selected Phe-Au complexes.

the carboxylate.

As a whole clusters with charge + 2 tend to form tridentate complexes with Phe^{-1}, whereas clusters with charge 1 + and 0 are able to form only bidentate complexes. Apparently, charge + 2 of cluster leads to more intense electrostatic bonding and should lead to a higher interaction energy.

3.3. Intracomplex Gibbs binding energies

We have evaluated the Gibbs binding energy for each complex with one to six gold atoms and charge −1, 0 and + 1. Overall, the highest binding energy is observed for the Au_3^{2+} cluster: the binding energy ΔG° equals 101 kcal mol^{-1} (Fig. 3). Among the cationic clusters, Au_2^{1+} is the most effective one: ΔG° equals 66.8 kcal mol^{-1}. Among the neutral clusters, Au_4^0 is the most stable one: ΔG° equals 38 kcal mol^{-1}. We also should mention the Au^{3+} cation which has an extremely large binding energy equal to 418.8 kcal mol^{-1}. Apparently, *in vitro* Au^{3+} cations should be absorbed on Phe prior to gold reduction and nanocluster formation [8]. Larger neutral gold clusters Au_8 and Au_{20} possess remarkable, however, moderate binding energies: 23.9 kcal mol^{-1} and 21.2 kcal mol^{-1}, respectively. Thus, smaller clusters form more energetically favorable complexes with Phe.

For the above mentioned complexes (Phe^{-1}- Au_3^{2+}, Phe^{-1}- Au_2^{1+} and Phe^{-1}- Au_4^0), we have performed quantum-topological analysis.

3.4. Bader's QTAIM analysis

Bader's quantum theory «Atoms-In-Molecules» allows to examine the wavefunction and perform its topological analysis. This gives valuable information on the nature of intraatomic interactions and bond energies. The Bader's study of the electronic density involves following attributes:

- all electrons density ρ(r);
- the electron density Laplacian $\nabla^2\rho(r)$;
- the kinetic energy Lagrangian G(r);
- the potential energy density V(r);
- the energy density E(r).

The methodology and interpretation of QTAIM analysis results have been published by us already, for details please see our previous papers [51,52]. All electron density Laplacians $\nabla^2\rho(r)$ are positive which means that all Au-X bonds are electrostatic. The values of energy density E(r) are all negative which means the stabilizing interactions of Au-X bonding. Finally, negative total energy E(r) values mean that bonds are partially electrostatic and partially covalent.

QTAIM analysis of bonds critical points has shown that Au-N interaction possesses the highest bond energy and predominates in the complex formation (Table 1). For example, in Phe^{-1}-Au_3^{2+} Au-N, Au-O and Au-C bond energies are equal to 62.9 kcal mol^{-1}, 55.2 kcal mol^{-1} and 37.4 kcal mol^{-1}, respectively. In the Phe^{-1}-Au_2^{1+} and Phe^{-1}- Au_4^0 complexes, Au-N bond predominates over Au-C bond also. Therefore, the bond energies decrease in the following order: Au-N > Au-O > Au-C.

The highest bond energy is observed for the Au-N interaction of the Phe^{-1}-Au_2^{1+} complex (70.7 kcal mol^{-1}). However, the summation of all intracomplex Au-X bonds gives a higher value for Phe^{-1}-Au_3^{2+} (155.5 kcal mol^{-1}) than for the Phe^{-1}- Au_2^{1+} (100.3 kcal mol^{-1}) and Phe^{-1}- Au_4^0 (97.3 kcal mol^{-1}). As a whole, the cumulative intracomplex bond energy decreases in the same order as the Gibbs binding energy Phe^{-1}- Au_3^{2+} > Phe^{-1}- Au_2^{1+} > Phe^{-1}- Au_4^0. Hence, generally the results of Bader's bond energy evaluation and Gibbs binding energy calculation are in agreement.

3.5. Simulation of UV–vis absorption spectra

Next, we have simulated the absorption spectra of the complexes using time-dependent density functional theory (TDDFT). Prior to TDDFT calculations, we have performed a small benchmark using a precise *ab initio* method similarity transformed equation of motion coupled-cluster method with single and double excitations (STEOM-CCSD). This method accurately predicts electronic absorption spectra for various systems including complicated metal–organic ones [53]. The UV spectrum of tetrachloroaurate ($AuCl_4^-$) possesses two maxima: at 225 nm and 315 nm [54]. STEOM-CCSD accurately predicts this spectrum with two peaks located at 229 and 322 nm. Among 10 DFT functionals (PBE, PBE0, M06, M062X, B3LYP, CAM-B3LYP, TPSS, TPSSh, WB97 and WB97X), M062X and CAM-B3LYP gave the best results: their maxima were at 227 nm and 331 nm, at 221 nm and 331 nm, respectively (Fig. S1). Moreover, M062X and CAM-B3LYP proved their efficacy predicting the spectra of transition metal–ligand complexes earlier [42,43].

Next, we have regarded one of the most stable Phe-Au systems, namely Phe^{-1}-Au_4. Its STEOM-CCSD spectrum has two major maxima: at 320 nm and 384 nm; M062X maxima occur at 329 nm and 390 nm, whereas CAM-B3LYP maxima are located at 325 nm and 373 nm. Therefore, both functionals nicely reproduce the spectra of a single Au^{3+} ion and the Phe-Au complex.

We have examined the spectra of two more Phe-Au systems using TDDFT. We established 20 vertical transitions for the most stable complexes among systems with charge + 1 (Phe^{-1}-Au_3^{2+}) and 0 (Phe^{-1}-Au_2^{1+}). First, Au_3^{2+} complex possesses long-wavelength transition $S_0 \rightarrow S_1$ located at 849 nm with a low oscillator strength equal to 0.002 according to M062X and 840 nm (0.005) according to CAM-B3LYP. The predominant transition of Phe^{-1}-Au_3^{2+} is located at 355 nm (Fig. 4) with $f_{OSC} = 0.151$ (361 nm, $f_{OSC} = 0.117$). Thus, the low intensity of the $S_0 \rightarrow S_1$ transition makes it doubtful for luminescent detection of the complex since according to Kasha's rule fluorescence should occur from the S_1 state and should also possess high intensity. The isolated Au_3^{2+} cluster possess the longwave maximum in the infra-red region at 1440 nm ($f_{OSC} = 0.002$) according to M062X and at 1259 nm (0.003) according to CAM-B3LYP. Thus, this transition undergoes dramatic hypsochromic shift upon complexation of Au_3^{2+} with Phe^{-1}. The predominant band of the bare Au_3^{2+} is located at 282 nm (0.294) according to M062X and at 298 nm (0.104) according to CAM-B3LYP. Therefore, this band bathochromically shifts upon the complexation of Au_3^{2+} cluster with Phe^{-1} (Fig. 4).

Phe^{-1}-Au_2^{1+} is another complex with a magic number of confined electrons (n* = 2). This system has a longwave maximum corresponding to a $S_0 \rightarrow S_1$ transition at 676 nm (0.015) and 817 nm (0.015) according to M062X and CAM-B3LYP, respectively. The predominant transition is located at 287 nm with $f_{OSC} = 0.092$ (304 nm, $f_{OSC} = 0.08$). The bare Au_2^{1+} cluster possesses a major maximum at 291 nm (0.554) and at 293 nm (0.436) according to M062X and CAM-B3LYP, respectively. The longwave maxima are located at 565 nm (0.009) and 520 nm (0.01), respectively. Thus, the longwave maximum of the Au_2^{1+} cluster undergoes a significant redshift upon complexation with Phe, whereas the most intense transition undergoes a dramatical loss of absorption intensity.

The predominant transition at 329 nm with $f_{OSC} = 0.893$ (325 nm, $f_{OSC} = 0.664$) of the Phe^{-1}- Au_4^0 complex is accompanied by a 390 nm longwave maximum with $f_{OSC} = 0.037$ (373 nm, $f_{OSC} = 0.052$), which relates to a $S_0 \rightarrow S_1$ transition. The UV–vis M062X spectrum of the isolated Au_4 cluster is similar to the CAM-B3LYP Au_4 spectrum (Fig. 4): major maximum is located at 364 according to M062X, at 366 nm according to CAM-B3LYP. The longwave maximum is located at 478 nm and 453 nm, respectively; it is hypsochromically shifted as a result of interaction with Phe.

Amiri-Sadeghan et al. reported circular dichroism (CD) spectra of Phe-Au NCs demonstrating that different functional groups drive different combinations of hydrogen bonding which is an important force in the self-assembling process of the molecules [8]. For this reason, we have simulated CD spectra of Phe^{-1}-Au complexes also (Fig. S2). Each CD

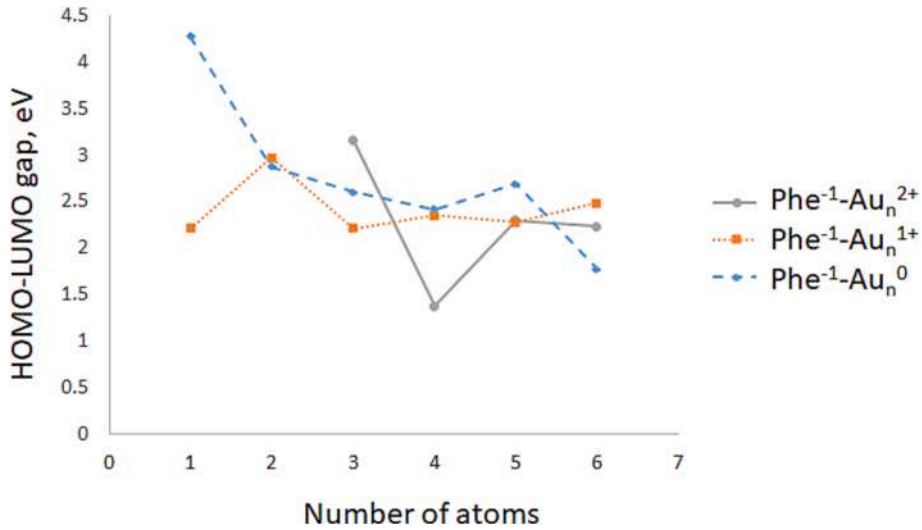

Fig. 5. The HOMO-LUMO energy gap of Phe-Au complexes depends on cluster size and charge.

spectrum can be described by a characteristic minimum: for example, Phe^{-1}-Au_3^{2+} minimum is located at 535 nm. Apparently, our CD spectra should be helpful to resolve the structure of experimental Phe-Au NCs.

Phe^{-1}-Au_4^0 and Phe^{-1}-Au_2^{+1} complexes are probably luminescent since their $S_0 \rightarrow S_1$ transitions have relatively high oscillator strengths. To estimate the sensitivity of the clusters to Phe we have also calculated the HOMO-LUMO energy gap depending on the cluster size and charge (Fig. 5). We found a trend: the HOMO-LUMO gap decreases with the growth of cluster size. For example, Phe^{-1}-Au_8 and Phe^{-1}-Au_{20} (not present on Fig. 5) possess HOMO-LUMO gaps equal to 1.5 eV and 1.48 eV, which is significantly lower than the gap of Phe^{-1}-Au_6 (1.77 eV). Also, the clusters with a magic number of confined electrons (Phe^{-1}-Au^0, Phe^{-1}-Au_2^{1+} and Phe^{-1}-Au_3^{2+}) possess the highest HOMO-LUMO gap. Apparently, upon the attachment of Phe, the spectra of these clusters undergo a hypsochromic shift (Fig. 4), whereas others, non-magic number clusters are characterized by a bathochromic shift of the electronic spectra.

3.6. Comparison with UV–vis spectra of Tyr-Au systems

Finally, to evaluate the selectivity of absorption spectra and Au clusters towards Phe^{-1} we have compared the spectra of Phe^{-1}-Au systems versus the complexes involving alanine and tyrosine (Fig. S3-S5). Tyrosine is also an aromatic amino acid similar to Phe, whereas alanine has no aromatic ring. Like in the case of Phe, we have used deprotonated alanine, deprotonated tyrosine Tyr^{-1}, and tyrosine semiquinone $SemiQ^{-1}$.

As one can see, all four Au_2^+ cluster complexes spectra drastically distinguish from each other (Fig. S3). Phe^{-1}-Au_2^+ possesses a characteristic longwave maximum at 678 nm ($f_{OSC} = 0.015$) and 818 nm ($f_{OSC} = 0.015$) according to M062X and CAM-B3LYP, respectively. Similar bands are possessed by Tyr^{-1}-Au_2^+ complex at 717 nm (0.037) and 911

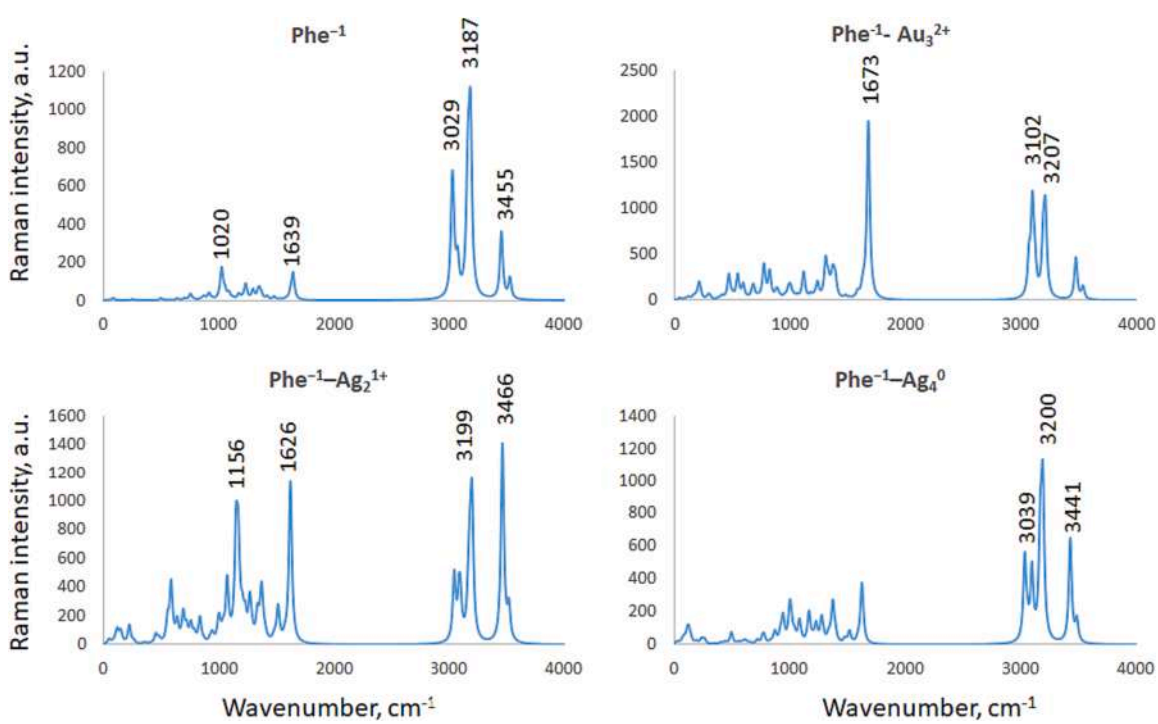

Fig. 6. Simulated Raman spectra of the major Phe-Au complexes.

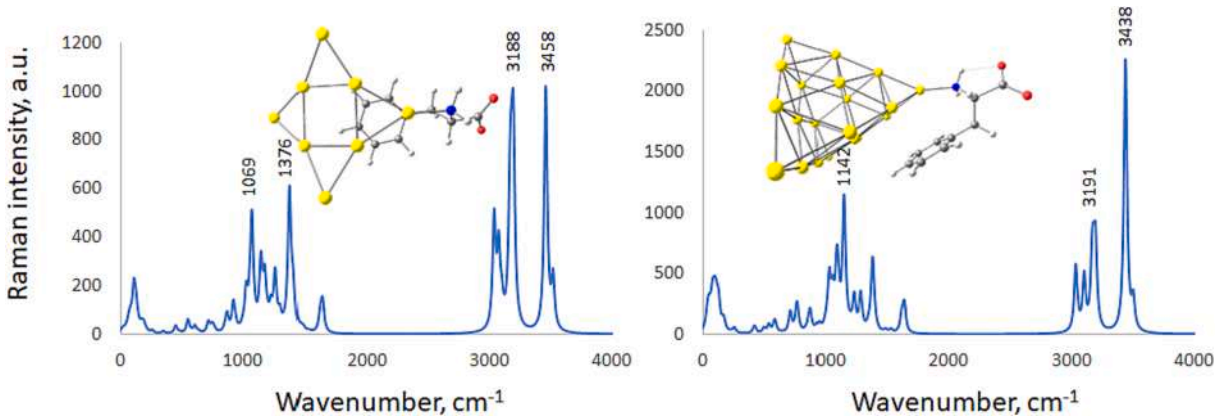

Fig. 7. Simulated Raman spectra and geometry of the Phe complexes with Au_8 and Au_{20} clusters.

(0.048) according to M062X and CAM-B3LYP, respectively. Nevertheless, these bands are much more pronounced and red-shifted as compared to Phe^{-1}-Au_2^+. The most significant transitions of Ala^{-1}-Au_2^+ and SemiQ^{-1}-Au_2^+ spectra are located in the UVB region,

Regarding Au_3^{2+} complexes (Fig. S4), the UV range of the spectra contain dominant transitions at the 300–350 nm range, whereas the visible part of the spectra possesses low oscillator strength and zero intensity transitions.

Au_4^0-containing systems are all similar: their optically bright transitions are located in the UV region and have similar character. The predominant transitions of Ala and Tyr are located above 350 nm, whereas Phe and SemiQ dominant transitions are below 350 nm. Thus, Au_2^{1+} cluster seem to be the most potent tool for UV–vis detection of Phe in amino acid mixtures. Moreover, as one can see open-shell systems have an extended UV–vis spectrum absorbing light even in the infra-red region, whereas closed-shell systems absorb light solely in the UV range.

3.7. Raman spectra of Phe-Au systems

First, we have examined the spectrum of the isolated Phe^{-1}, and then we have analyzed how the addition of gold changes the Raman spectrum. The spectrum of the bare Phe^{-1} has two significant peaks at the 3000–3500 cm^{-1} area (Fig. 6). The most dominant one is located at 3187 cm^{-1} (1119 a.u.), it corresponds to the C–H stretchings of the phenyl ring. The next significant peak is at 3029 cm^{-1} (678 a.u.) and relates to the symmetric stretching of the β-CH_2 group. The 3455 cm^{-1} band (355 a.u.) corresponds to the symmetric stretching of the amino group.

The phenyl ring angular bending vibrations as in-phase C–C–C motions of the phenyl ring are well-known in literature and occur at 1004 cm^{-1} and 1016 cm^{-1} according to experiment and theory, respectively [28]. According to our calculations, this is a 1020 cm^{-1} transition. The phenyl ring bond-stretching vibrations, assigned to the C–C in-phase motion is responsible for the 1606 cm^{-1} and 1644 cm^{-1} bands according to the experiment and theory, respectively. We have observed this transition at 1639 cm^{-1}. As one can see from these examples, our calculations are in agreement with the experiment and previous studies. The data on Phe-Au Raman spectra reported here have not been published previously.

The major transition in the Raman spectrum of Phe^{-1}-Au_3^{2+} is located at 1673 cm^{-1} (1933 a.u.) and corresponds to the COO^- group stretching. A couple of intense transitions is located at 3000–3500 cm^{-1} range. The first one at 3102 cm^{-1} (1179 a.u.) relates to the α-C–H bond stretching,

whereas the second one at 3207 cm^{-1} (1127 a.u.) corresponds to the C–H stretching of the phenyl ring.

The spectrum of the Phe^{-1}-Au_2^{1+} complex has four significant bands. The most intense one is located at 3466 cm^{-1} (1399 a.u.), it corresponds to the symmetric stretching of the amino group. The 3199 cm^{-1} (1159 a.u.) band next to it is similar to the 3207 cm^{-1} band of Phe^{-1}-Au_3^{2+} (Fig. 6): it also corresponds to the C–H stretching of the phenyl ring. The 1626 cm^{-1} band (1143 a.u.) is located at the center of the spectrum and related to the in-phase C–C motion of the phenyl ring. The next band at 1156 cm^{-1} (981 a.u.) involves two transitions: at 1147 cm^{-1} (β-CH_2 group twisting) and 1165 cm^{-1} (wagging of the amino group).

The two most intense bands of the Phe^{-1}-Au_4^0 spectrum are located at 3200 cm^{-1} (1139 a.u.) and 3441 cm^{-1} (648 a.u.). The first band corresponds to the C–H stretching of the phenyl ring, whereas the second peak is a symmetric stretching of the amino group. The 3039 cm^{-1} (569 a.u.) and 3102 cm^{-1} (505 a.u.) bands correspond to the symmetric and antisymmetric stretching of the β-CH_2 group, respectively.

We have simulated the infrared (IR) spectra of the bare Phe and major Phe-Au complexes (Fig. S6). The spectrum of the isolated Phe^{-1} is characterized by a predominant peak located at 1579 cm^{-1} (874 a.u.); it relates to the asymmetric stretching of the COO^- group. The experimental peak of the bare Phe at alkaline pH is located at 1574 cm^{-1} [55]. The major maximum of Phe^{-1}-Au_4 is located at 1618 cm^{-1} (1040 a.u.) and also corresponds to the asymmetric COO^- group stretching. This is in full agreement with the experimental data on IR spectra of Phe on Au (1 1 1): the 1616 cm^{-1} band corresponds to the asymmetric C1-O stretching of the carboxyl [55]. Next to it, stands another pronounced peak at 1373 cm^{-1} (547 a.u.); it relates to the symmetric COO^- group stretching, whereas the experimental band is located at 1410 cm^{-1} [55]. Thus, computational IR spectra are in agreement with experimental IR spectra of Phe-Au complexes. The predominant peak of the Phe-Au_2^+ complex is located at 1624 cm^{-1} (1116 a.u.), and also corresponds to the asymmetric stretchings of the COO^- group. Next to it, stands an intense maximum at 1371 cm^{-1} (648 a.u.; symmetric COO^- group stretching). Thus, our computational IR spectra are in agreement with the experimental ones.

Since complexes of larger clusters with a size n > 20 are exploited as models for SERS spectra [56], we have also simulated Raman spectra of Phe^{-1}-Au_8 and Phe^{-1}-Au_{20} complexes (Fig. 7). Phe^{-1}-Au_8 spectrum did not reveal significant enhancement of the Raman signal. The 3000–3600 cm^{-1} region has two predominant maxima at 3188 cm^{-1} (1004 a.u.) and 3458 cm^{-1} (1004 a.u.), which correspond to phenyl C–H symmetric stretching and symmetric stretching of the amino group, respectively. In the 1000–1600 cm^{-1} region, the major maxima are at 1069 cm^{-1} (510 a.u.) and 1376 cm^{-1} (598 a.u.), which relate to αC-NH_2 bond stretching and αC-βC-H scissoring, respectively.

The Phe^{-1}-Au_{20} complex spectrum possesses a predominant peak at 3438 cm^{-1} (2263 a.u.) responsible for symmetric stretchings of the amino group; therefore, the enhancement of Raman signal is > six times as compared to the naked Phe^{-1}. Next to him, stands a 3191 cm^{-1} maximum (917 a.u.), which corresponds to C–H stretching of the phenyl ring. The 1000–1600 cm^{-1} region is significant for the 1142 cm^{-1} (1072 a.u.) transition (wagging of the amino group). Thus, ultrasmall clusters

Table 2
Results of natural bond orbital analysis for the Phe-Au systems.

Complex	Donor (i)	Orbital Type	Occupancy	Acceptor (j)	Orbital Type	Occupancy	$E^{(2)}$, kcal/mol	q_{CT}
Phe^{-1}–Au_3^{2+}	N	n	0.906	Au1	n*	0.351	18.99	0.152
Phe^{-1}–Au_3^{2+}	C8-C9	π	0.812	Au3	n*	0.350	23.21	0.196
Phe^{-1}–Au_2^{1+}	N	n	0.884	Au1	n*	0.047	10.97	0.038
Phe^{-1}–Au_2^{1+}	N	n	0. 884	Au1-Au2	σ*	0.085	16.25	0.068
Phe^{-1}–Au_4^{0}	N	n	1.752	Au1	n*	0.731	63.01	0.212

and larger Au_8, Au_{20} systems possess similar Raman spectra and the level of chemical enhancement of the signal.

3.8. Natural bond orbital analysis

We have used second order perturbation theory analysis of Fock matrix in the NBO basis to establish the most significant intracomplex charge transfers. NBO analysis gives significant data when investigating intracomplex bonding and interaction between bonds. Electron donor and acceptor orbitals occupancies and the interaction stabilization energy ($E^{(2)}$) resulting from the B3LYP-D3(BJ) NBO analysis are demonstrated in Table 2. The larger the value of stabilization energy, the more intense the interplay between the electron donor orbital *i* and the electron withdrawing orbital *j* demonstrating a stabilizing donor–acceptor interaction. In the case of Phe and gold clusters the most intense intracomplex interactions are formed by the orbital overlap between *n*, π, *n**, Ry* and π* orbitals. We have already published the detailed NBO analysis of the interactions between metal clusters and similar aromatic amino acid tyrosine [51].

We have established the most pronounced intracomplex ground state charge transfers (CTs) for Phe^{-1}-Au_3^{2+}: $n_N \rightarrow n^*_{Au1}$ and $\pi_{C8\text{-}C9} \rightarrow n^*_{Au3}$ (for atom numbering see Fig. 2; for NBO orbitals see ESI Fig. S7). The latter CT could be responsible for the 3207 cm^{-1} band in the Raman spectrum of the complex. For $E^{(2)}$ energy and q_{CT} see Table 2.

Two strong (>10 kcal mol^{-1}) ground state CTs are observed for the Phe^{-1}-Au_2^{1+} system (Fig. S8). The first one is a $n_N \rightarrow n^*_{Au1}$ CT with $E^{(2)}$ equal to 10.97 kcal mol^{-1} and $q_{CT} = 0.038$. The second one is a $n_N \rightarrow n^*_{Au1\text{-}Au2}$ CT with $E^{(2)} = 16.25$ kcal mol^{-1} and $q_{CT} = 0.068$. These CTs could be responsible for the 3199 cm^{-1} and 1156 cm^{-1} bands in the Raman spectrum of Phe^{-1}–Au_2^{1+}.

The Phe^{-1}-Au_4^0 complex possesses a strong CT $n_N \rightarrow n^*_{Au1}$ with a high $E^{(2)}$ equal to 63.01 kcal mol^{-1} (Fig. S9). These N → Au1 CT could be responsible for the 3441 cm^{-1} band (648 a.u.) in the Raman spectrum of Phe^{-1}–Au_4^0 and symmetric stretching of the amino group.

Therefore, the main transitions in the regarded Phe–Au systems are ligand–metal (Table 2), primarily amino group-donating transitions. Apparently, the ligand–metal transitions are prospective for Raman spectroscopy since they lead to a significant chemical enhancement in the spectra. Thus, a combination of the NBO analysis and the ground state CT evaluation gives reliable and well interpretable results.

4. Conclusion

We have analyzed the interactions of phenylalanine with ultra-small gold clusters with charge 0–2 and a size up to six atoms, as well as neutral Au_8 and Au_{20} clusters. The most stable complexes (with the lowest Gibbs binding energy) have been selected for further analysis. We have performed the UV–vis absorption and Raman spectra simulation. Also, we have performed QTAIM and NBO analysis of the intracomplex interactions.

Au_2^{1+} is the most probable for luminescent detection of Phe since $S_0 \rightarrow S_1$ transition of its complex with Phe at 356 nm has a relatively high oscillator strength. Also, Phe^{-1}-Au_2^{1+} complex allows to distinguish Phe from Tyr and tyrosine semiquinone. Au_{20} cluster seems to be the most prospective for Raman detection of Phe since it leads to nearly six-times enhancement of the Raman signal. Thus, exploitation of the regarded and larger gold clusters and cluster-involving interfaces should be beneficial for the detection of Phe in biological fluids. For example, this can be done using dual-emitting Au clusters with high selectivity and sensitivity, which can be used for luminescent detection of Phe [57]. This highly selective nanosensor acts through the interplay between Phe and surface motifs of Au NC. Phe promotes the charge transfer effect through the ligand-core interface thus enhancing fluorescence emission intensity.

Bader's QTAIM theory showed that metal–ligand intracomplex interactions are partially covalent and partially electrostatic. The NBO analysis have shown that the main transitions in the regarded Phe-Au systems are ligand → metal and occur with the participation of the amino group.

CRediT authorship contribution statement

Andrey A. Buglak: Writing – original draft, Visualization, Investigation, Funding acquisition, Formal analysis, Conceptualization. **Alexei I. Kononov:** Writing – review & editing, Supervision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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Appendix A. Supplementary data

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